Solving Tensor Linear Systems under the t-product by a Kaczmarz-type method

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Why Tensors?

- Images, videos, and multi-dimensional data are naturally represented as **third-order tensors** $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$.
- **Image reconstruction**: given a blurred/noisy observation $\mathcal{B}$, recover the clean image $\mathcal{X}$ from

$$\mathcal{A} * \mathcal{X} = \mathcal{B}$$

  where $*$ denotes the *t-product*.
- Systems are often **large-scale** and **inconsistent**.

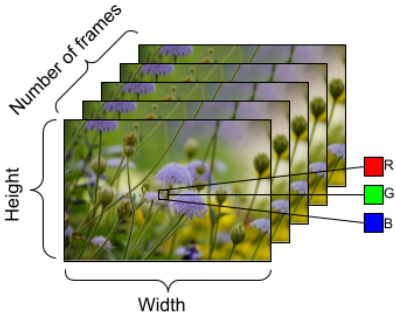
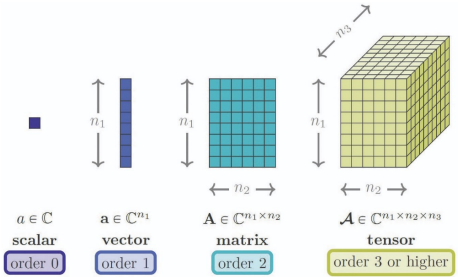


Image as a 3D tensor (H x W x C)



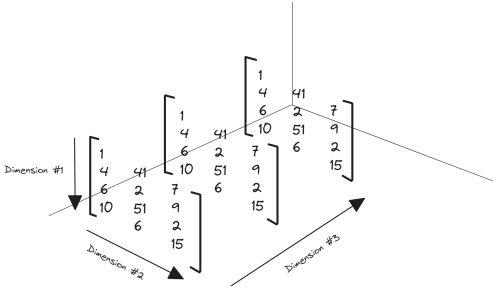


Figure: Illustration of third-order tensors.

The t-Product: Why and How

Definition (Kilmer & Martin, 2011): Given $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $\mathcal{X} \in \mathbb{R}^{n \times k \times p}$, the t-product $\mathcal{A} * \mathcal{X}$ is:

$$\mathcal{A} * \mathcal{X} \;=\; \mathrm{fold}\Big(\mathrm{bcirc}(\mathcal{A}) \cdot \mathrm{unfold}(\mathcal{X})\Big) \;\in\; \mathbb{R}^{m \times k \times p}$$

where $\mathrm{bcirc}(\mathcal{A}) \in \mathbb{R}^{mp \times np}$ is the block-circulant matrix of $\mathcal{A}$, and $\mathrm{unfold}(\mathcal{X}) \in \mathbb{R}^{nk \times p}$ is the mode-3 unfolding of $\mathcal{X}$, defined as:

$$\operatorname{bcirc}(\mathcal{A}) := \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_{N_3} & \cdots & \mathbf{A}_2 \\ \mathbf{A}_2 & \mathbf{A}_1 & \cdots & \mathbf{A}_3 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{N_3} & \mathbf{A}_{N_3-1} & \cdots & \mathbf{A}_1 \end{bmatrix} \in \mathbb{R}^{N_1 N_3 \times N_2 N_3}.$$

$$\operatorname{unfold}(\mathcal{A}) := \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_{N_3} \end{bmatrix} \in \mathbb{R}^{N_1 N_3 \times N_2}, \qquad \operatorname{fold}\left(\begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_{N_3} \end{bmatrix} \right) := \mathcal{A}.$$

Why the t-product?

- Classical matrix algebra cannot directly capture the **multi-channel structure** of tensors (e.g. RGB channels, video frames).
- The t-product defines a **matrix-like algebra for third-order tensors**, preserving notions of transpose, inverse, SVD, and rank.

Key properties:

- Efficient computation via **FFT along the tube dimension**
- Supports transpose ($\mathcal{A}^\top$), pseudoinverse ($\mathcal{A}^\dagger$), and t-SVD
- Enables a full **linear algebra framework** for 3D data

Cyclic and Randomized Kaczmarz

Consider a linear system $Ax = b$ with rows $a_i^\top$ of A.

Kaczmarz update (row projection):

$$x^{k+1} = x^k + \frac{b_{i_k} - a_{i_k}^\top x^k}{\|a_i\|_2^2} a_{i_k}$$

- At each step, project x^k onto the hyperplane $\{x : a_i^\top x = b_i\}$.
- Cyclic version scans rows in order $i_k = k(\mod i) + 1$

Randomized Kaczmarz (RK):

- Pick row i_k randomly, with $\Pr(i_k = i) = \|a_i\|_2^2 / \|A\|_F^2$.
- Same projection formula with $i = i_k$.
- For consistent and overdetermined systems, RK converges *linearly in expectation* and satisfies:

$$\mathbb{E}[\|x^k - x^\star\|_2^2] \leq \left(1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2}\right)^k \|x^0 - x^\star\|_2^2$$

- For inconsistent systems $Ax = b + \epsilon$, the **randomized extended Kaczmarz (REK)** variant applies doubly the RK method to the systems

$$Ax = b - z \quad A^\top z = 0$$

.

- converges linearly in expectation to the minimum-norm *least-squares solution*.

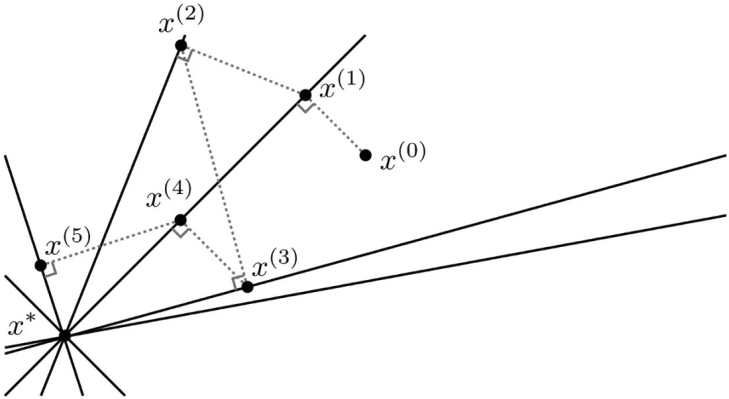


Figure: Illustration of Kaczmarz iterations as projections onto hyperplanes defined by the rows of A.

Problem Setup

Tensor Linear System

Given $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $\mathcal{B} \in \mathbb{R}^{m \times k \times p}$, find $\mathcal{X} \in \mathbb{R}^{n \times k \times p}$ satisfying:

$$\mathcal{A} * \mathcal{X} = \mathcal{B} + \epsilon$$

Goal: compute the *minimum-norm least-squares* solution

$$\mathcal{X}^\star = \underset{\mathcal{X}}{\operatorname{argmin}} \|\mathcal{X}\|_F \quad \text{s.t.} \quad \mathcal{X} \in \underset{\mathcal{Z}}{\operatorname{argmin}} \|\mathcal{A} * \mathcal{Z} - \mathcal{B}\|_F$$

Setting:

- System is **inconsistent** ($\mathcal{B} \notin \mathrm{range}(\mathcal{A})$)
- Dimensions m, n large $\Rightarrow$ direct solvers impractical
- Iterative Kaczmarz-type methods are well-suited

Existing Methods & Limitations

Tensor Kaczmarz variants in the literature:

- **TREK** – Tensor Randomized Extended Kaczmarz [7]
- **TREBK** – Tensor Randomized Extended Block Kaczmarz [7]
- **TREGBK** – Tensor Randomized Extended Greedy Block Kaczmarz [7]
- **TREABK** – Tensor Randomized Extended Averaged Block Kaczmarz [2, 4]

Key limitation:

Pre-defined Partitions Required

TREBK, TREGBK, and similar block methods require a **pre-defined partition** of the row/column indices.

- Partition quality directly affects convergence
- No principled way to choose the partition a priori
- Extra tuning overhead

TGDBEK: Key Idea

Two-step greedy selection at each iteration k:

Z-step (column selection):

- Score each column slice by its contribution
- Greedily select the set U_k of most informative columns
- Project: $\mathcal{Z}^{k+1} \leftarrow \mathcal{Z}^k - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^k$

X-step (row selection):

- Score each row slice by its residual
- Greedily select the set J_k of most informative rows
- Update: $\mathcal{X}^{k+1} \leftarrow \mathcal{X}^k + (\mathcal{A}_{J_k,:,:})^\dagger * \mathcal{R}_{J_k}$

No pre-defined partitions needed!

Algorithm 1: Tensor Greedy Double Block Extended Kaczmarz (TGDBEK)

Input: $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$, $\mathcal{B} \in \mathbb{R}^{N_1 \times K \times N_3}$, iterations ℓ, parameter $\eta \in (0, 1]$.

Initialize: $\mathcal{X}^{(0)} = \mathcal{O}$ and $\mathcal{Z}^{(0)} = \mathcal{B}$.

for $k = 0, 1, \ldots, \ell - 1$ **do**

$$\varepsilon_k^z = \eta \max_{1 \le j \le N_2} \left\{ \frac{\left\| (\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)} \right\|_F^2}{\left\| \mathcal{A}_{:,j,:} \right\|_F^2} \right\}$$

$$U_k = \left\{ j \in [N_2] : \left\| (\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)} \right\|_F^2 \ge \varepsilon_k^z \left\| \mathcal{A}_{:,j,:} \right\|_F^2 \right\}$$

Update $\mathcal{Z}^{(k+1)} = \mathcal{Z}^{(k)} - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^{(k)}$

$$\varepsilon_k^x = \eta \max_{1 \le i \le N_1} \left\{ \frac{\left\| \mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)} \right\|_F^2}{\left\| \mathcal{A}_{i,:,:} \right\|_F^2} \right\}$$

$$J_k = \left\{ i \in [N_1] : \left\| \mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)} \right\|_F^2 \ge \varepsilon_k^x \left\| \mathcal{A}_{i,:,:} \right\|_F^2 \right\}$$

Update $\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + (\mathcal{A}(J_k,:,:))^\dagger * \left(\mathcal{B}_{J_k,:,:} - \mathcal{Z}_{J_k,:,:}^{(k+1)} - \mathcal{A}(J_k,:,:) * \mathcal{X}^{(k)} \right)$

end

The TGDBEK Algorithm

Greedy Selection Criterion

Given a parameter $\eta \in (0, 1]$:

Column selection (Z-step):

$$\varepsilon_z^k = \eta \cdot \max_j \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^k\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2}, \quad U_k = \left\{ j : \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^k\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \geq \varepsilon_z^k \right\}$$

Row selection (X-step):

$$\varepsilon_x^k = \eta \cdot \max_i \frac{\|\mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{k+1} - \mathcal{A}_{i,:,:} * \mathcal{X}'^k\|_F^2}{\|\mathcal{A}_{i,:,:}\|_F^2}, \quad J_k = \left\{ i : \text{score}_i \geq \varepsilon_x^k \right\}$$

- $\eta \to 1$: fewer, most informative slices (aggressive)
- $\eta \to 0$: more slices selected (conservative)
- U_k, J_k are **determined adaptively** at each iteration

Convergence Guarantee

Theorem

Let $\mathcal{A} \in \mathbb{R}^{m\times n\times p}$ and $\mathcal{B} \in \mathbb{R}^{m\times k\times p}$. let $\mathcal{B}^{\perp} := \mathcal{B} - \mathcal{A} * \mathcal{A}^{\dagger} * \mathcal{B}$. The iterate $\{\mathcal{X}^{(k)}\}$ generated by TGDBEK algorithm converges linearly to the minimum-norm least-square solution $\mathcal{X}_* = \mathcal{A}^{\dagger} * \mathcal{B}$. Moreover, for $k \geq 1$, the following holds

$$\|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 \leq \alpha^{k+1}\|\mathcal{X}_0 - \mathcal{X}_*\|_F^2 + \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta}\|\mathcal{B}^{\perp}\|_F^2$$

where $\alpha = 1 - \theta\frac{\sigma_{\min}^2(\mathrm{bcirc}(\mathcal{A}^{\top}))}{\sigma_{\max}^2(\mathrm{bcirc}(\mathcal{A}^{\top}))}$, $\beta = 1 - \frac{\sigma_{\max}^2(\mathcal{A})}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}(\overline{U})\|_F^2}$, $\gamma = \sigma_{\min}^2(\mathcal{A}^{\mathcal{J}})$, $\theta \in (0, 1]$ *and* $\eta \in (0, 1]$.

- *Full proof available in the technical report.*

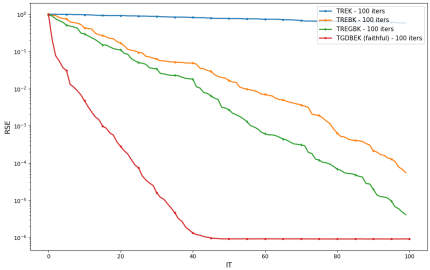
Results on Sparse Matrices (SuiteSparse)

ash85 (85×85, density $= 0.072$)

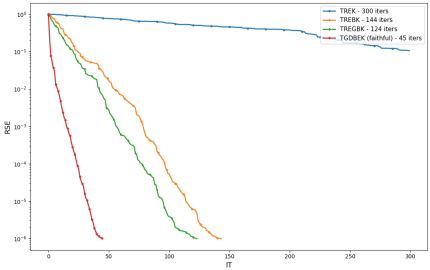
Method	**Time**	**RSE**	**IT**
TREK	0.370	2.0e−2	500
TREBK	0.369	9.2e−7	174
TREGBK	0.574	9.9e−7	258
TGDBEK	**0.248**	9.3e−7	**78**

nos5 (153×153, density $= 0.047$)

Method	**Time**	**RSE**	**IT**
TREK	0.454	1.0e−1	300
TREBK	0.860	9.8e−7	144
TREGBK	0.864	9.9e−7	124
TGDBEK	**0.581**	9.7e−7	**45**

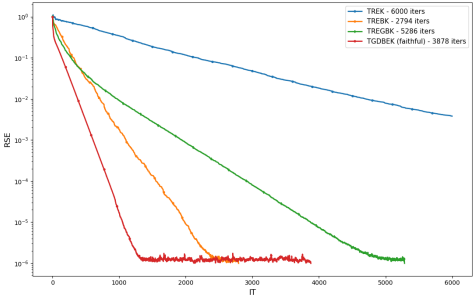


RSE vs. Iterations (ash85)



Cities Matrix (128×128), density $= 0.072$)

Method	**Time**	**RSE**	**IT**
TREK	4.089	3.8e−3	6000
TREBK	3.318	9.9e−7	2794
TREGBK	6.943	9.9e−7	5286
TGDBEK	5.954	9.9e−7	**3878**



RSE vs. Iterations (Cities)

Takeaway:

- TGDBEK consistently reaches target accuracy with **fewer iterations**
- Competitive or better wall-clock time **without needing partition tuning**

Results on Color Image Deblurring & Reconstruction

Color image deblurring: TGDBEK achieves higher PSNR and faster convergence than TREK/TREBK/TREGBK on standard test images.

Original image

Blurred image

Summary & Future Work

Summary:

1. **No pre-defined partitions**: greedy selection adapts at each step
2. **Faster convergence** : fewer iterations than TREK, TREBK, TREGBK
3. **Proven guarantees** : linear convergence to min-norm LS solution

Experiments on Image reconstruction & Color image deblurring

Future work:

- Larger-scale experiments (high-resolution images, video)
- GPU-accelerated implementation via PyTorch
- Extensions to other tensor decompositions (Tucker, TT)
- Applications to tensor completion and compressed sensing

Thank you!

Code:
https://github.com/jnlandu/tensor-greedy-double-extended-kaczmarz

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